NMC 2010 O-LEVEL PAPER 2 QUESTIONS
SECTION A

Qn.1. If $\frac{a}{b+c+d} = \frac{4}{3}$ and $\frac{a}{b+c} = \frac{3}{5}$. What is the value of $\frac{a}{a}$?

Qn.2. Alice and Bob play a game involving a circle whose circumference is divided into 12 equally spaced points. The points are numbered clockwise from 1 to 12. Both start on point 12. Alice moves clockwise and Bob moves counterclockwise, In a turn of the game, Alice moves 5 points and Bob moves 9 points. The game ends when they stop on the same point. How many turns will this take?

Qn.3. If $1 + x + x^2 + x^3 = 0$, What is the value of $1 + x + x^2 + x^3 + \ldots + x^{2010}$?

Qn.4. Suppose that $f(x) = ax + b$, given that

$$f(f(f(x))) = 8x + 21$$

Determine the value of $a + b$.

Qn.5. George's company was losing money, as a result George received a pay cut of 25%. By what percentage must his new pay rate be raised to bring it back to the original level?

Qn.6. Find the value of x if

$$\left(\sqrt{3}\right)^x \cdot \left(\sqrt{27}\right)^x = \sqrt{3}$$

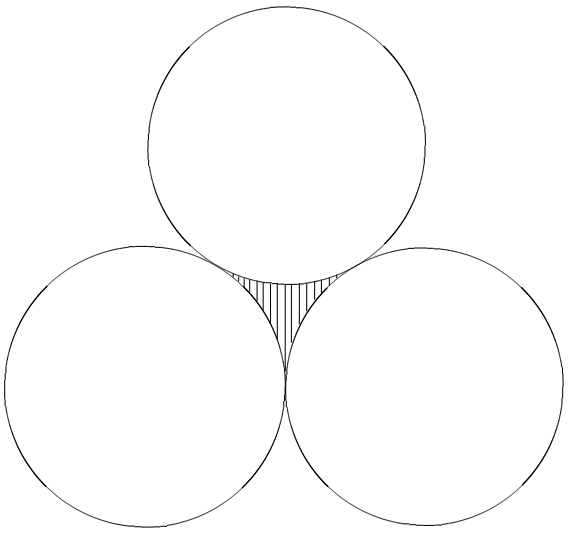
Qn.7. Find the value of $2x + 3y$ given that $x + y = 6$ and $x^2 + 3xy + 2y^2 = 60$.

Qn.8. Find the sum and product of all the solutions to the equation

$$2\log_{10} x - \log_{10} 2x = 1.$$

Qn.9. Mark had an average score of 85 on his first ten quizzes, and 90 on his fourteen quizzes in total. What is the average score of his last 4 quizzes?

Qn.10. Three circles of radius 1 meter are mutually tangent as shown in the figure. What is the area of the shaded region?



SECTION B : 10 **marks each**

Qn.11. (a) If the letters a, A, b, B, c and C are arranged at random in a row. What is the probability that the lower case letters appear in an increasing alphabetical order?

(b) Suppose a rectangle has an area of 3 and a diagonal of length $\sqrt{10}$. What is it's perimeter?

Qn.12. James and Ann leave a camp site. James biking due north and Ann biking due east. James bikes $7km/hr$ faster than Ann. After three hours, they are $51km$ apart. Find the sum of their speeds.

Qn.13. Juma can run around a circular track in 40 seconds, David, running in the opposite direction meets Juma every 15 seconds. What is the average of the times David and Juma take to run around the track, expressed in seconds.

Qn.14. A student on vacation for d days observed the following

- it rained 7 times, morning or afternoon
- when it rained in the afternoon, it was clear in the morning
- there were 5 clear afternoons
- there were 6 clear mornings

Using this information, find the number of days the student was on vacation?

Qn.15. In the figure below, $\overline{AB} = \overline{AC}$, angle $BAD = 30^\circ$, and $\overline{AE} = \overline{AD}$, find the value of the angle EDC.

